

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE/IT) Examination

Nov-Dec 2013

IT 206/IT 206.01 Object Oriented Programming using Java

Date: 12.11.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the Following Questions.

[07]

Fill in the blanks.

- 1) The Java interpreter translates a Java program from _____ to machine language.
- 2) A (n) _____ occurs when a program has a grammatical mistake like missing punctuation.
- 3) If the computer prints an error message while executing your program, the program contains a (n) _____.
- 4) The value of $42 \% 10$ is _____.

State true or false.

- 5) An upper-case letter is considered to be the same character as a lower-case letter in Java.
- 6) The value of $5 / 2$ and the value of $5.0 / 2$ are the same.
- 7) What will be the value of x after the following section of code executes:

```
int x = 5;  
if (x > 3) { x = x - 2; }  
else { x = x + 2; }
```

Q - 2 Answer the Following Questions.

- (A) Which two conditions should be met so that an automatic type conversion will take place? Explain widening conversion and narrowing conversion. [06]
- (B) Write down the differences between method overloading and method overriding with example. [04]
- (C) What is static block? What are static variables? Mention the difference between static and non-static variables? [04]

OR

Q - 2 Answer the Following Questions.

- (A) List the logical operators are available in java. Explain how the short circuit logical operators (&&, ||) useful in programming? [06]
- (B) Explain the different forms of Polymorphism by taking suitable example. [04]
- (C) Which types of access specifiers available in Java? Explain how final modifier can be affect to declaration of class, method and data? [04]

Q - 3 Answer the following Questions.

- (A) Explain Life cycle of an applet in detail. [05]
- (B) What is an interface? Can we instantiate an interface? Which kind of member variables do interfaces have? [05]
- (C) Write a short note on : Event handling using AWT [04]

OR

Q - 3 Answer the following Questions.

- (A) Explain Life Cycle of Thread in detail. [05]
- (B) How are this () and super () used with constructors? Explain the multiple and multi-level inheritance. Can super () be a last statement in constructor? Why? [05]
- (C) Write a short note on : Types of classes in java [04]

SECTION – II

Q - 4 Answer the Following Questions.

[07]

Fill in the blanks:

- 1) Compiling a file called Game.java will produce a file called _____.
- 2) To create an object of a class type, the word _____ is used.
- 3) A (n) _____ is a collection of related classes that serves as a class library.
- 4) An array that uses more than one index is called a (n) _____ array.

State true or false:

- 5) The size of an array in Java is determined when the program is compiled.
- 6) An equals method should return a boolean value.
- 7) What will the following code print?

```
int count = 1;
while (count <= 3)
{   System.out.print(count + " ");
    count++; }
```

Q - 5 Answer the following Questions.

- (A) Write an application that creates ten Rock objects and saves these in an array. Randomly select mass between 1 and 10 kilograms for each rock as it is created. After all rocks have been created, display their individual masses and total mass of all rocks. [06]
- (B) Differentiate the following. [06]
 - 1) Event vs. State
 - 2) Composition vs. Aggregation
 - 3) Generalization vs. Association
- (C) Explain Fork, Join concept. [02]

OR

Q - 5 Answer the following Questions.

- (A) Write an application that illustrates how a method can invoke a superclass method. Class J2 is extended by K2. Class K2 is extended by L2. Each of these classes defines a getDescription () method that returns a string. That string includes a description of the class plus description of each superclass. Instantiate each of these classes and invoke the getDescription () method. [06]
- (B) Draw use case diagram for Railway Reservation system. [06]
- (C) List out the different components of state chart diagram. [02]

Q - 6 Answer the following Questions.

- (A) Draw state chart diagram for Telephone Calling System. [07]
- (B) Explain OMT Model. [07]

OR

Q - 6 Answer the following Questions.

- (A) Draw Activity Diagram for Order Placing System. [07]
- (B) Draw the use case diagram for ATM System. [07]
